PriceWise AI

Requirements Document

Software Construction and Development - Assignment 1

1. Project and Group Information

Project Title	PriceWise AI
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Course	Software Construction and Development

2. Problem Description

Online shoppers often struggle to know whether they are getting the best possible price for a product, since prices for the same item vary across different retailers and change frequently. Manually checking multiple stores for every purchase is time-consuming, and shoppers frequently overpay simply because they only checked one or two sources. PriceWise AI addresses this problem by automatically searching multiple online stores for a given product, comparing their prices in one place, and suggesting similar, cheaper alternatives when available.

3. Project Objectives

- Allow users to search for a product and view prices from multiple online retailers in one place.
- Automatically highlight the lowest available price for a searched product.
- Recommend similar, lower-priced alternative products based on category and specification overlap.
- Track a product's price over time and visualize price history.
- Allow users to save products and set target-price alerts, notifying them when a price drops.
- Provide secure user authentication and a personal dashboard for saved products and alerts.

4. Intended Users

- Online shoppers looking to compare prices before purchasing.
- Budget-conscious buyers and students looking for the best deal.
- Anyone who wants to track a product's price over time before deciding to buy.

5. Functional Requirements

1. The system shall allow a user to create an account using email and password, or sign in using Google.
2. The system shall allow a logged-in user to search for a product by name.
3. The system shall retrieve and display prices for the searched product from multiple online stores.
4. The system shall automatically identify and highlight the lowest available price among the results.
5. The system shall suggest alternative products that are similar to the searched product but lower in price.
6. The system shall allow a user to save a product to a personal wishlist.
7. The system shall allow a user to set a target price for a product and be notified by email when the price drops to or below that target.
8. The system shall record and display a product's price history over time.
9. The system shall maintain a record of a user's past searches.

10. The system shall provide a dashboard where a user can view their wishlist, active price alerts, and search history.

6. Non-Functional Requirements

- The application should return search results within a few seconds under normal conditions.
- The application should be usable on both desktop and mobile screen sizes.
- User passwords must be stored securely using password hashing, not plain text.
- The system should handle a failed or slow external price data request gracefully, without crashing.
- Access to user-specific data (wishlist, alerts, history) must be restricted to the authenticated owner of that data.